

Strategies for Establishing Clinical-Decision Thresholds in Psychiatry

A Review

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IMPORTANCE Nearly all psychopathology is inherently continuous, but both clinical practice and research often require categorical decisions (eg, treat or wait, pay for treatment or decline, enroll in study or not); therefore, thresholds are applied to continuous phenomena to guide these decisions. The most common strategy is to select cutoffs by focusing on diagnoses, but this may be unhelpful in many decisions (eg, selective prevention, choice between treatment options). This review evaluates strategies for developing thresholds for psychopathology dimensions that may enhance clinical decision-making, drawing on methods used in psychiatry, psychology, internal medicine, and health economics.

OBSERVATIONS One strategy used extensively in neuropsychology, child psychiatry, clinical psychology, and laboratory medicine is to define multiple thresholds (eg, mild, moderate, and severe), using statistical deviance vis-à-vis the general population. A second strategy is to select thresholds in reference to functional impairment, as has been done for some psychopathology measures. A third approach, common in internal medicine but underused in psychiatry, references probability of a negative outcome (eg, mortality). A fourth option, gaining momentum in European psychiatry, is to select thresholds based on costs and benefits of a clinical action at different levels of severity or risk. Common examples of these 4 strategies are reviewed, comparing their strengths and limitations. They are relevant not only to clinical measures but also research instruments. Value judgments are inherent in approaches 2, 3, and 4; thus, selection of such thresholds requires stakeholder input. Importantly, clinicians make decisions by considering all information, and thresholds are only 1 factor.

CONCLUSIONS AND RELEVANCE Although diagnosis-focused thresholds are sometimes valuable, the field would be served best by also considering the aforementioned alternative strategies. These approaches have been greatly underused in psychiatry, and studies are urgently needed to address this gap. Herein, recommendations are offered for this research.

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Extensive evidence indicates that most forms of psychopathology are distributed continuously across the general population.¹⁻⁷ Likewise, genetic vulnerabilities to mental disorders are primarily continuous.⁸ To date, no mental disorder has been established as a natural category.⁹ Hence, psychopathology may be similar to cardiovascular and metabolic disorders that fall on a continuum with normal physiological functioning, such as essential hypertension with normal blood pressure and type 2 diabetes with blood glucose.^{10,11} This continuity is emphasized by alternative nosologic models including an extended psychosis phenotype,¹² clinical staging models,^{13,14} Research Domain Criteria,¹⁵ and Hierarchical Taxonomy of Psychopathology.¹⁶

Nevertheless, categorical decisions are often necessary in clinical practice (eg, treat or wait, hospitalize or not) and research (eg, meets study inclusion criteria or not), which require continu-

ous psychopathology dimensions to be categorized. For example, a Patient Health Questionnaire 9 score greater than 20 is an indication for combined pharmacotherapy and psychotherapy.¹⁷ Health care professionals use such thresholds in conjunction with other relevant information to make clinical decisions. Optimally, severity cutoffs should be tailored empirically to particular decisions.^{18,19} For instance, low cutoffs would be optimal for administering preventive interventions, higher ones for treatment initiation, and still higher ones for intensive interventions (eg, hospitalization), thus identifying multiple actionable ranges on a dimension. However, the strategy for the development of ranges has not been explicitly articulated in psychiatry. Goals of this review are to contrast the widespread diagnosis-focused strategy with 4 alternative approaches for threshold selection. We present evidence that the resulting ranges can make psychopathology assessment more accurate and useful.

Figure 1. Examples of Thresholds Established Using Statistical Deviance: Personality, Psychopathology, and Cognitive Ability Measures

A Personality or psychopathology measures

Personality or psychopathology measures	T Scores						
	50	55	60	65	70	75	
	50	70	85	Percentiles 93	97.5	99	
ASEBA	Normal				Borderline	Clinical	
PROMIS: adults	Normal		Mild	Moderate		Severe	
IDAS	Normal		Mild		Moderate		Severe
PAI	Normal	Moderately elevated				Markedly elevated	
SNAP-2	Normal		Moderately elevated		Elevated		
MMPI-3	Average		Moderately elevated		Elevated		Extreme

B Cognitive ability measures

	T Scores				
Cognitive ability measures	100	90	80	70	
	50	25	Percentiles 9	2.5	
WISC-V	Average		Low average	Borderline	Extremely low
WAIS-IV, WIPPSI-IV	Average		Low average	Borderline	Extremely low
WJ-IV COG	Average		Low average	Borderline	Extremely low

ASEBA indicates Achenbach System of Evidence-based Assessment²⁹; IDAS, Inventory of Depression and Anxiety Symptoms³³; MMPI-3, Minnesota Multiphasic Personality Inventory 3rd edition³²; PAI, Personality Assessment Inventory³⁰; PROMIS, Patient-Reported Outcomes Measurement Information System³⁴; ranges are provided for the PROMIS Emotional Distress measures; SNAP-2, Schedule for Nonadaptive and Adaptive Personality-2nd edition³¹; WAIS-IV, Wechsler Intelligence Scale for Adults 4th edition²⁷; WIPPSI-IV, Wechsler Preschool and Primary Scale of Intelligence 4th edition²⁵; WISC-V, Wechsler Intelligence Scale for Children 5th edition²⁶; WJ-IV COG, Woodcock-Johnson IV Tests of Cognitive Abilities.²⁸

Diagnosis-Focused Thresholds

The current research practice for threshold selection relies heavily on aligning the cutoff with the diagnosis. Specifically, thresholds on continuous measures are typically chosen to maximize sensitivity and specificity for detection of the target disorder.^{20,21} Resulting cutoffs are then used for various questions that require dichotomization without adjustment or validation.

Diagnoses themselves reflect continua that were dichotomized, such as the number of symptoms (eg, at least 5 of 9 criteria) and duration of disturbance (eg, ≥ 2 weeks). Diagnostic thresholds have been chosen mainly following heuristics and expert clinical judgment.²² These semiarbitrary rules are ingrained in traditional diagnoses. They are unavoidable, as no natural boundaries have been established in psychopathology.^{1-7,9} Accordingly, cutoffs selected based on diagnoses inherit semiarbitrary decisions that defined these disorders. Another limitation is that such thresholds are tailored to 1 application (ie, disorder screening) and are not optimal for others (eg, prognosis or treatment selection).^{18,19,22} We next discuss approaches to address these limitations.

Statistical-Deviance Approach

Thresholds can be selected based on the distribution of scores in the general population. Ranges can be defined by extremity and quantified as a percentile or standard score.¹⁸ Indeed, statistical deviance is routinely used in internal medicine when interpreting results of biochemical analyses of blood and tissue.²³ The common definition of the reference interval (ie, normal range) is the central 95% of a healthy population, meaning that a test result above the

97.5th percentile is considered elevated and below the 2.5th percentile is considered decreased. Moderate and severe elevations can be characterized by even greater statistical deviance. These elevations can guide treatment decisions. For instance, 95% of male individuals have a blood hemoglobin concentration greater than 13.0 g/dL (to convert to grams per liter, multiply by 10), and lower values are an indication for clinical action.²⁴

Many psychological assessments have adapted thresholds based on statistical deviance. This strategy is near-universal among cognitive tests²⁵⁻²⁸ and dimensional measures of child psychopathology.²⁹ It is also standard for personality pathology inventories^{30,31} and symptoms scales for adults.³²⁻³⁴ Statistically based ranges for 10 widely used measures are given in Figure 1. These instruments differ in threshold number, boundaries, and labels. Such ranges are inherently conventions, given lack of natural discontinuities. Nevertheless, a consistent pattern is notable, with moderate elevation typically defined as a T score greater than 65 (top 7% of the reference population), marked elevation as a T score greater than 70 (top 2.5%), and severe elevation as a T score greater than 75 (top 1%). Cognitive assessments use similar thresholds but focus on low scores (eg, bottom 2.5%). These ranges allow continuous information to be summarized in a few simplified rubrics (eg, normal, mild, moderate, marked), but graded information is also available when more precision is needed.

Overall, the statistical-deviance strategy is a feasible option and has proven its utility in various fields. Importantly, ranges should be derived in representative samples and updated regularly to maintain validity. Also, statistical deviance is only a partial solution, as the resulting ranges do not inform specific clinical actions. Alternatively, threshold can be selected based on external criteria that define actions (eg, determining eligibility for a service). Next, we review 3 such strategies.

Table 1. Examples of Thresholds Linked to Impairment

Measure	Sample	Thresholds	Method
IDAS-II ³⁸ Spanish version	Adults >18 y Community sample Patient sample	Set for 19 IDAS-II scales: 5.5–43.5 for moderate impairment 6.5–48.5 for severe impairment	- IDAS-II scales were linked to WHODAS 2.0 (Spanish version) impairment ranges via ROC >25 WHODAS 2.0 total score: moderate impairment >50 WHODAS 2.0 total score: severe impairment - Thresholds were selected based on the best balance between sensitivity and specificity
Ham-D ₁₇ ³⁹ Spanish version	Adults >18 y Patient sample	≤5 for Normal functioning	- HAMD17 total score was linked to the SOFAS (Spanish version) threshold for normal functioning ≥80 via ROC - Threshold was selected to maximize both sensitivity and specificity
Ham-D ₁₇ ⁴⁰ Japanese version	Adult >18 y Patient sample	≤4 for Remission	- HAMD17 total score was linked to health-related quality of life indexed by SF-36 (Japanese version) of remitters vs nonremitters via ROC and logistic multivariate analyses - Threshold was selected to maximize AUC or achieve maximal level of agreement with health-related quality of life recovery
ADHD ⁴¹ DSM-IV	18–75 y	>4 Symptoms HI >4 Symptoms IA	- ROC analyses; criterion was a composite of impairment items (eg, interference with social contacts and activities) dichotomized at 95th percentile in the general population - Thresholds were selected to maximize the sum of sensitivity and specificity
CSI ⁴²	Adults >18 y Medicaid beneficiaries	>30 Clinical range	- ROC analyses; criterion was qualifying for disability associated with psychiatric disorder - Threshold was selected to maximize sensitivity and specificity
AUDIT ⁴³	Adults >18 y	8–10	Alcohol-related problems (domestic, legal, and occupational difficulties coded as present/absent) detected with maximal sensitivity and specificity via ROC analyses

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; AUDIT, Alcohol Use Disorders Identification Test; CSI, Colorado Symptom Index; HAMD17, 17-item Hamilton Rating Scale for Depression; HI, hyperactive subtype of ADHD; IA, inattentive subtype of ADHD; IDAS-II, Inventory for Depression and Anxiety Symptoms II; ROC, receiver operating characteristic curve; SF-36, Medical Outcome Study 36-Item Short-Form Health Survey scale used to evaluate health-related quality of life; SOFAS, Social and Occupational Functioning Assessment Scale; WHODAS 2.0, World Health Organization Disability Assessment Scale 2.0.

Functional-Impairment Approach

Impaired functioning in society (ie, social, educational, occupational, etc) may be the single most important criterion for psychopathology. It is part of the clinical significance criterion that *DSM-5-TR*³⁵ uses to distinguish mental disorder from problems in living. This criterion helps to determine eligibility for services and whether individuals meet administrative and legal requirements (eg, reimbursement, reporting). Clinical significance reflects a philosophical position that mental disorder is *harmful dysfunction*,³⁶ with harm being necessary for symptoms to be considered pathological. Symptoms reflect problems with mental functions (eg, emotional responses, arousal, cognition), whereas functioning in society is their real-world consequences. Distress is the second factor in the clinical significance criterion, but its entanglement with symptoms³⁷—especially depression and anxiety—makes distress a weak external criterion.

Accordingly, functional impairment is a reasonable criterion for threshold selection. Table 1 provides examples of such thresholds developed for several widely used instruments.^{38–43} These studies frequently used receiver operating characteristic (ROC) analysis, as it compares sensitivity and specificity across different cutoffs. Some studies identified multiple psychopathology ranges by linking them to different levels of impairment.^{38–40}

This research faced 2 challenges. First, functioning is multifaceted, and it may not be obvious on which aspect to focus. Many thresholding studies addressed this issue using global

indices, eg, total scores on the Social and Occupational Functioning Assessment Scale (SOFAS)⁴⁴ and World Health Organization (WHO) Disability Assessment Schedule 2.0 (WHODAS 2.0).^{38,39} The downside of such criteria is that they are heterogeneous, with a given global score representing different impairment profiles, which makes it harder to connect the resulting threshold to a specific clinical action. A few studies examined specific criteria such as alcohol-related problems or qualifying for psychiatric disability.^{42,43}

Second, most functional metrics are naturally continuous. If relationships between symptoms and functioning were nonlinear, then cutoffs could be placed at a point where the association strengthens. However, existing evidence largely supports linear associations,^{5,6} and no nonlinear effects have been replicated. Therefore, functioning itself must be dichotomized for use as a criterion. Some studies dichotomize functioning according to statistical deviance (eg, bottom 5th percentile).⁴¹ Others use measures with explicit anchors for levels of impairment. For example, the SOFAS⁴⁴ consists of 10 tiers, mild, moderate, and severe, permitting its dichotomization by desired impairment level. Nevertheless, this raises questions about the rigor with which anchors were designed. For example, SOFAS⁴⁴ is based on the Health-Sickness Rating Scale,⁴⁵ which includes descriptive anchors derived through consensus of 9 clinicians. Ultimately, categories of impairment are conventions identified rationally or via statistical deviance. Conventions are not necessarily problematic, but they define applicability for the resulting thresholds. For instance, a cutoff selected in reference to SOFAS less than or equal to 30

Table 2. Examples of Thresholds Linked to Future Outcomes

Measure	Sample	Thresholds	Method
Examples from internal medicine			
SBP/DBP ⁵⁰	Age, 20-74 y		Risk of stroke/heart disease relative to normal: (HR)
		Normal (SBP: <120-129 mm Hg; DBP: <80 mm Hg)	1
		Elevated (SBP: 120-129 mm Hg; DBP: <80 mm Hg)	1.1- to 1.5-Fold increase
		Stage 1 hypertension (SBP: 130-139 mm Hg; DBP: 80-89 mm Hg)	1.5- to 2.0-Fold increase
		Stage 2 hypertension (SBP: >140 mm Hg; DBP: >90 mm Hg)	>2.0-Fold increase
Total cholesterol, mg/dL ⁵¹	40-69 Males		Risk of cardiovascular death relative to desirable: (HR)
		Desirable: <200	1
		Borderline high: 200-239	1.1
		High: ≥240	2.96
HbA _{1c} , % ⁵²	Age, 45-79 y		% Risk of developing diabetes relative to reference: (HR)
		Reference (5.0-5.5)	1
		Prediabetic (6.0 to 6.5)	6.29-Fold increase
BMI ^{53a}	Age, 20-74 y		Risk of death over 10 y relative to normal: (RR)
		Underweight (<18)	1.38- to 2.30-Fold increase
		Normal (18.5-24.9)	1
		Overweight (25-29.9)	No increase in risk
		Class 1 (30-34.9)	1.13- to 1.20-Fold increase
		Classes 2 & 3 (≥35)	1.63- to 1.83-Fold increase
PCE ^b for CVD, % ⁵⁴	Age, 40-79 y		% Risk for developing CVD in 10 y relative to low: (RR)
		Low (0-4.9)	1
		Borderline (5-7.4)	2.0- to 3.0-Fold increase
		Intermediate (7.5-20)	3.0- to 8.0-Fold increase
		High (>20)	>8.0-Fold increase
Examples from psychiatry			
Beck Hopelessness Scale ⁵⁵	Age, 15-84 y Outpatients	High-risk (≥9)	• ROC; criterion was suicide at follow-up (1-7 y later) • 94% Sensitivity; 41% specificity • Maximized sensitivity
Beck Depression Scale ⁵⁵	Age, 15-84 y Outpatients	High-risk (≥23)	• ROC; criterion was suicide at follow-up (1-7 y later) • 77% Sensitivity; 62% specificity • Maximized sensitivity
Scale for Suicide Ideation-Worst ⁵⁶	Adolescent and adult outpatients	Higher risk (≥16)	• ROC; criterion was suicide at follow-up (4-15 y) • 80% Sensitivity; 78% specificity • Maximized both sensitivity and specificity
SNAP-2 Self Harm ⁵⁷	Age, 18-45 y Outpatients	Sensitive cutoff (≥10) Optimal cutoff (≥12)	• ROC; criterion was suicide attempt in 1 y • 84% Sensitivity; 70% specificity • 72% Sensitivity; 85% specificity
AUDIT ⁵⁸	Adult hospital patients	High risk (≥8)	• ROC; criteria: social problems, trauma, related medical disorder, alcohol-related treatment 2-3 y later • Maximized both sensitivity and specificity across various outcomes
Nine commonly used assessment tools for risk of future violent, criminal, and sexual offenses ⁵⁹	73 Samples 13 Countries Adults with a history of violent criminal offenses	Thresholds indicating high/moderate and low risk on the nine commonly used assessment tools for risk of reoffending	• Odds ratios; ROC; criteria: risk of violent, criminal, and sexual offending • Areas under the curve 0.66-0.74 • Summary odds ratio for future violent outcomes = 6.1 • 92% Sensitivity; 36% specificity for violent outcomes
NAPLS-2 risk calculator ⁶⁰	596 Clinical high-risk participants Mean age, 18.5 y	Threshold = 0.4 of model-predicted risk	• Time to event proportional hazards regression • Probability of 2-y conversion to psychosis = 16% • Harrell C index = .71 • Maximal positive predictive value = 48.4% • Specificity = 96.9%

Abbreviation: AUDIT, Alcohol Use Disorders Identification Test; BMI, body mass index; CVD, cardiovascular disease; DBP, diastolic blood pressure; HbA_{1c}, hemoglobin A_{1c}; HR, hazard ratio; NAPLS-2, second phase of the North American Prodrome Longitudinal Study; PCE, pooled cohort equations; ROC, receiver operating characteristic curve; RR, relative risk; SBP, systolic blood pressure; SNAP-2, Schedule for Nonadaptive and Adaptive Personality-2nd

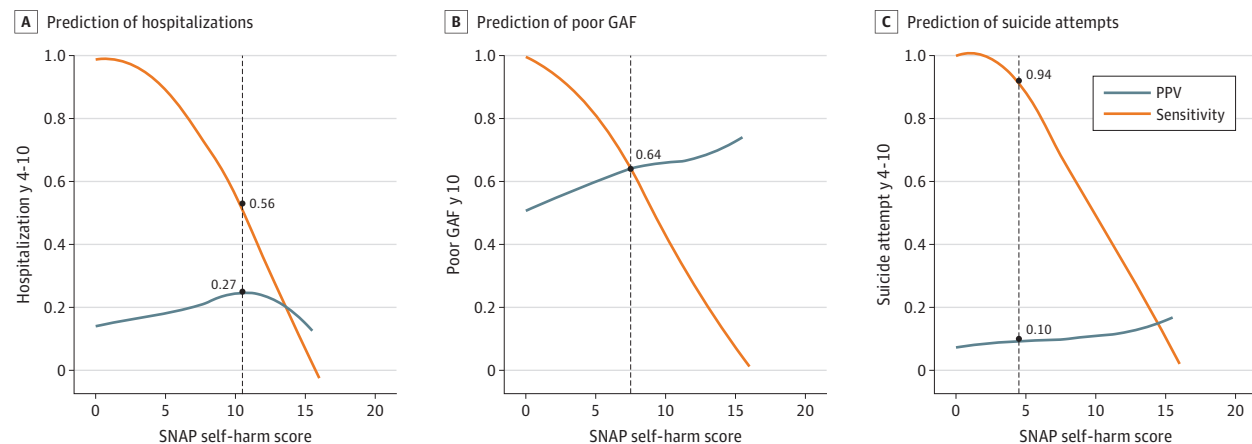
edition.

SI conversion factor: To convert total cholesterol to millimoles per liter, multiply by 0.0259; HbA_{1c} to proportion of total hemoglobin, multiply by 0.01.

^a Calculated as weight in kilograms divided by height in meters squared.

^b PCE for CVD risk = % of risk for developing CVD in 10 y.

Figure 2. Line Plots Showing Changes in Positive Predictive Value (PPV) and Sensitivity Across Cutoffs



To illustrate development of risk ranges, we conducted secondary analyses of Collaborative Longitudinal Personality Disorders Study (CLPS; described in detail elsewhere).⁶³ We used data from 430 patients with psychiatric disorders, such as personality disorders or major depressive disorder and no personality disorder, who had outcomes data. Risk factor of interest was Schedule for Nonadaptive and Adaptive Personality-2nd edition (SNAP-2) Self-Harm, which measures propensity to suicide including self-loathing and self-destructive behaviors.⁵⁷ Its scores range from 0 to 16. SNAP Self-Harm was assessed at baseline. Outcomes were assessed 10 years later, including suicide attempts, hospitalizations for psychiatric disorders, and Global Assessment of Functioning (GAF).⁴⁴ These outcomes were selected because self-harm is relevant to each. To minimize overlap between predictor and outcome, outcomes were assessed years after baseline. GAF was rated for the best 6 months between years 8 and 10; suicide attempts and hospitalizations were counted only between years 4 and 10. The 2 latter outcomes are naturally dichotomous, but GAF ranges from 1 to 100. We defined poor outcome as GAF ≤ 60 , because these scores indicate illness severity that is at least moderate. Other cutoffs are equally defensible (eg, ≤ 50 , serious illness or ≤ 70 , any psychopathology vs none) and may be preferred in some studies. Poor GAF was common (prevalence of 50%), hospitalization was rare (14%), and suicide attempt was very rare (7%). We evaluated overall predictive accuracy of SNAP Self-Harm for each outcome. Area under the curve (AUC) analyses produced AUC of 0.68 for hospitalization, 0.68 for poor GAF, and 0.66 for suicide attempt, indicating moderate accuracy. Next, we compared thresholds possible on SNAP Self-Harm. Specificity, sensitivity, negative predictive value, PPV, and others can be used for this comparison. We were most interested in practical utility and thus examined PPV (ie, among patients identified by threshold, what proportion will experience the outcome) and sensitivity (ie, among patients who will experience the outcome, what proportion was identified by the threshold).⁶⁴ Low thresholds produce

high sensitivity but poor PPV, whereas high thresholds maximize PPV at the expense of sensitivity, so we had to consider relative importance of these statistics. For hospitalization, we sought to prioritize PPV, as this outcome was rare and most cutoffs produced PPVs that were too low to be practical. A, Maximum PPV was 0.25 and was found at SNAP Self-Harm score of >11 . This threshold had sensitivity of 0.55, which was acceptable. Therefore, cutoff >11 selected 23% of the sample; 25% of them will be hospitalized, and conversely, 55% of people who will be hospitalized were detected. This modest accuracy underscores the challenge of predicting low probability events. For poor GAF, PPV was much higher; therefore, sensitivity and PPV were weighed equally. B, We found that for cutoff >7 both of these statistics were equal at 0.64. In other words, this threshold selected 53% of the sample; 64% of them will have poor outcome, and 64% of poor outcome cases were detected. C, For suicide attempts, even maximal PPV was too low to identify cases in need of preventive intervention; therefore, this was approached as a negative screen, that is, a test to rule out patients not at risk. We deemed sensitivity of $>90\%$ to be necessary for such an application. Cutoff >4 had a sensitivity of 0.94 and PPV of 0.10 and identified 26% of the sample as low risk. This threshold can be used to screen out patients with high confidence as not needing suicide prevention intervention but captures only a portion of patients not at risk. In summary, scores 0-4 were low risk (screen negative for suicide attempts), 5-7 were normal risk typical of outpatients, 8-11 were moderate risk (poor outcome is likely), and 12-16 were highest risk (elevated odds of hospitalization). Each range was linked to a different outcome, but it is also possible to define ranges according to likelihood of a single outcome (Table 2). After ranges are defined on one scale, it is possible to apply them to other measures of the construct (eg, other suicidality scales), using equipercentile linking, rather than collecting longitudinal data for each measure.

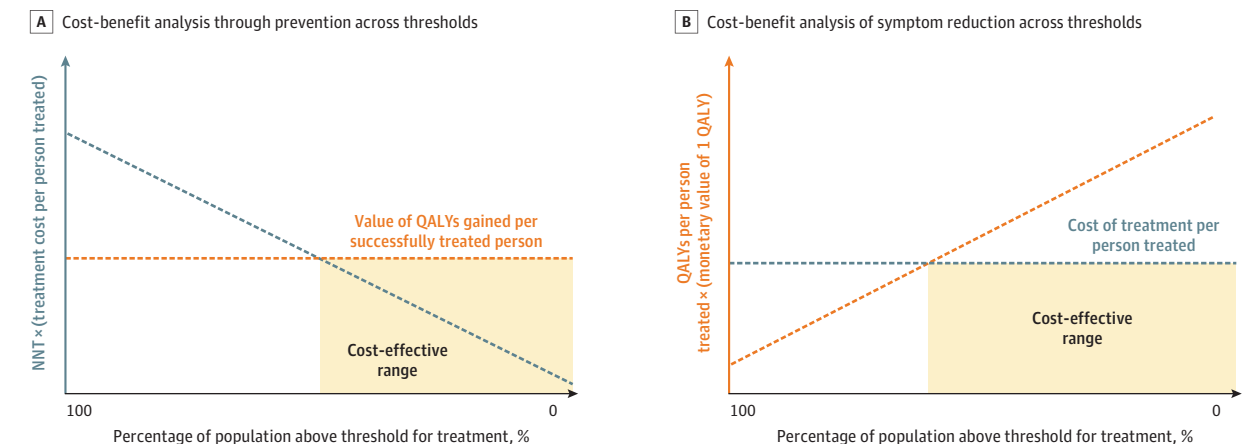
may be appropriate for decisions about hospitalization but not about initiation of psychotherapy.⁴⁶

Further, appropriateness of the functional impairment criterion for defining psychopathology is debated.⁴⁷ Indeed, the WHO separates health conditions, as described in *International Classification of Diseases, 11th Revision (ICD-11)*,⁴⁸ from associated disability, which is characterized in another manual, the *International Classification of Functioning, Disability, and Health (ICF)*.⁴⁹ Moreover, functional impairment is limited to current status and does not anticipate future negative consequences that may result from psychopathology. For instance, a person who gambles excessively may have avoided social consequences so far but is at increased risk of future interpersonal and financial problems. Cutoffs based on current functioning would miss such cases. This limitation can be addressed by thresholds developed to predict future outcomes.

Prediction-of-Future-Outcomes Approach

Internal medicine provides numerous examples of thresholds applied to dimensions to predict future outcomes. Table 2 summarizes several prominent examples, including biological measures (eg, body mass index, blood pressure, cholesterol, hemoglobin A_{1c}), and cardiovascular disease risk composite.⁵⁰⁻⁶⁰ These markers are considered fundamentally continuous and related to outcomes in graded fashion.^{50,52} Ranges are identified to guide clinical actions (eg, regular monitoring, lifestyle change, or pharmacotherapy) in reference to relevant outcome (eg, mortality risk, diabetes, or cardiovascular disease onset). Outcome risk can be expressed as hazard ratios (HRs) for the identified range compared with the reference interval (ie, normal range).

Figure 3. Conceptual Illustration of Threshold Selection Using the Cost-Benefit Approach



A, Analysis of poor outcome prevention. B, Analysis of burden reduction. Quantities given in parentheses remain constant across thresholds. NNT indicates number needed to treat. QALY, quality-adjusted life-years.

Ranges are usually labeled according to HR as elevated or borderline, intermediate, high, or very high. Notably, even modest HRs (ie, 10%-50% increase in risk) are often sufficient to designate the range as actionable. Although essential in internal medicine, the aforementioned risk factors are only moderately predictive. For example, the Pooled Cohort Equations (PCEs) is an aggregate of 8 best-established cardiovascular disease risk factors and predicts disease onset with an area under the curve (AUC) of 0.71 to 0.82.⁶¹

Limited examples of this approach exist for psychopathology (Table 2). Most research has focused on suicide attempts and completed suicide, evaluating various dimensional predictors.⁵⁵⁻⁵⁷ Some studies developed thresholds for prediction of future violent behavior⁵⁹ or onset of psychosis.⁶⁰ Others addressed thresholding of alcohol-related symptom severity vis-à-vis risk of negative outcomes.⁵⁸ Psychopathology studies have not calculated HRs but instead conducted ROC analyses, selecting thresholds that maximize either sensitivity or sensitivity plus specificity. Sensitivity (rate of false-negative results among people who will have the outcome) is emphasized when risk factors are used for screening, and hence, the primary concern is false-negative results. Sensitivity plus specificity—called *optimal cutoff*—is used when the goal is maximizing overall accuracy. Optimal cutoff assumes that false-positive results and false-negative results are equally important, although that is rarely the case.⁶²

It is useful to outline considerations in developing prediction-based cutoffs, as they remain rare for psychopathology despite availability of requisite longitudinal data, especially from youth and adult patient populations.^{19,63} Figure 2 illustrates a process that may be followed in such secondary analyses.⁶⁴ Specifically, we identified 4 ranges of risk (low, normal, moderate, and high) on the Self-Harm scale of the Schedule for Nonadaptive and Adaptive Personality–2nd edition (SNAP-2).³¹ We defined these ranges by analyzing outcomes of different severity: poor global outcome, hospitalization, and suicide attempt. We weighed sensitivity and positive predictive value (PPV) differently depending on the intended use of the range, such as sensitivity, which should

be maximized for screening and PPV maximized for initiation of intervention. In particular, we determined that 26% of outpatients had low risk of suicide attempts. The corresponding threshold provides a highly sensitive screen and narrows the population, but above-threshold scores do not necessarily indicate that suicide prevention intervention is needed. Instead, this cutoff can play a valuable role in sequential screening, where patients who screened positive undergo further evaluation to identify a smaller high-risk group for intervention.

SNAP-2 Self-Harm was only modestly accurate in predicting target outcomes, but it produced AUCs that were similar to AUCs of widely used cardiometabolic risk factors.^{52,61} These factors guide preventive care and public health policy, which has resulted in major increases in life expectancy. It is likely that risk factors with even modest accuracy (AUC approximately 0.70) will be similarly useful in psychiatry. Notably, SNAP-2 Self-Harm is a self-report instrument that requires only 2 minutes to complete and thus is highly scalable. However, risk screening is only 1 aspect of suicide prevention and not sufficient by itself.⁶⁵

Another consideration for predictive ranges is that statistical evidence alone is insufficient for cutoff selection. Value must be assigned to false-positive results and false-negative results, which requires judgment of experts and stakeholders. Internal medicine has addressed this issue by regularly convening expert panels to update ranges based on the newest evidence.^{50-52,54} Unfortunately, no such committees have been established in mental health so far. Importantly, resulting ranges indicate level of risk absent intervention but are silent about optimal treatment choice. Thresholds linked to treatment cost-effectiveness can address this gap.

Cost-Benefit-of-Interventions Approach

The most direct question for a clinical threshold is for whom a given intervention is indicated. It is important for treatment decisions and certain preventive interventions. Also, cost-effectiveness of alternative interventions can be a major factor in policymaking, with

cheaper interventions preferred if treatments are equally effective.^{66,67} These questions can be answered by comparing costs and benefits associated with target treatment at different levels of the psychopathology dimension.

The severity range within which benefits of intervention outweigh its cost can be identified using economic evaluation.⁶⁸ This type of analysis usually estimates the costs of intervention in dollars and its benefits in quality-adjusted life-years (QALYs; ie, healthier years gained due to reduction of morbidity and mortality). The cost of adverse effects can be included by assigning them a monetary value. Likewise, monetary value can be assigned to QALYs (eg, \$50 000 per QALY) to account for personal suffering, impact on family, and social cost (eg, lost productivity) associated with illness. Importantly, the value varies across populations, and its assignment requires judgment of experts and stakeholders.⁶⁸ Cost-effectiveness is expected to improve with higher thresholds.

In prevention of discrete outcomes (eg, disorder onset, suicide attempt), low thresholds define a group that includes true-positive results (ie, people who will have the outcome if untreated) but also many false-positive results (ie, those not actually at risk). Accordingly, number needed to treat (NNT) to prevent 1 outcome is large. In contrast, high thresholds result in a greater proportion of true-positive results and, thus, smaller NNT. NNT can be multiplied by cost of intervention to calculate cost per successfully treated person at a given threshold. QALYs per 1 outcome prevented can be estimated from epidemiologic studies. After monetary value is assigned to a QALY for the target population, we can identify a threshold beyond which the benefit (QALY converted to dollars) outweighs intervention costs per outcome prevented and, thus, defines the cost-effective range (Figure 3A).

The same logic applies to continuous outcomes, such as reduction in symptoms or impairment. We can use average improvement after treatment initiation across a given time interval (eg, 5 years) for people above a severity threshold to calculate QALYs gained. Average pretreatment symptoms are lower for lenient thresholds, which limits improvement per person, whereas higher thresholds allow larger improvement and greater QALYs. Cost-effective range is where the value of QALYs gained per person treated exceeds the cost of that intervention (Figure 3B).

Empirical evidence supports these explanations. Economic evaluation of preventive interventions for cardiometabolic diseases consistently found that, for a given treatment, cost per QALY gained gradually decreases (ie, cost-effectiveness increases) along the continuum of risk. This was observed for β -blockers, statins, coronary artery bypass, antihypertensives, metformin, insulin, and lifestyle modification treatments.^{61,69,70} For example, increasing prevention threshold on hemoglobin A_{1c} from 5.5% to 6.0% decreases cost per QALY gained from \$70 000 to \$24 000, even for a low-cost treatment.⁶¹

Studies of psychopathology have not conducted full economic evaluation of alternative thresholds, but some have examined effect of threshold on intervention efficacy. For example, efficacy of early educational intervention to improve children's IQ differs markedly across risk levels from NNT of 50 at lowest to 3 at highest risk.⁷¹ Likewise, a meta-analysis⁷² showed that computer-assisted cognitive therapy is more efficacious for patients with high vs low pretreatment depression (treatment

effect size Hedges $g = 0.87$ vs 0.40). Also, a large trial of alcohol-dependence intervention reported greater benefits for patients with severe vs mild pretreatment dependence (approximately 2-fold difference in benefit).⁷³ However, this evidence remains preliminary.

Some interventions may become ineffective at the highest levels of severity, thus setting a limit on the corresponding range. However, systematic evidence for this possibility does not yet exist. The largest limitation of the cost-benefit approach is a paucity of economic evaluations for psychopathology thresholds. This gap can be addressed by secondary analyses that stratify data from published randomized clinical trials and epidemiologic studies of illness course by initial symptom severity or risk factor level. These results should be integrated with estimates of treatment costs and disability weights, which are readily available in the literature.⁷⁴ Accordingly, cost-benefit thresholds can be identified for many psychopathology dimensions using existing datasets. However, this approach to mental health services remains largely untested.

Implications and Recommendations

Thresholds developed in reference to diagnoses dominate the literature. However, they are tied to semiarbitrary criteria, as mental disorders lack natural boundaries,⁹ and tailored to just 1 question (ie, has the disorder or not); therefore, thresholds are likely not appropriate for various other applications. We reviewed 4 alternative strategies for cutoff selection, each with its strengths and weaknesses. First, statistical-deviance thresholds capture distribution of scores in the general population, summarizing it with standard rubrics (eg, significant elevation is top 7%, marked is top 2.5%, and severe is top 1%). This approach relies on feasible studies and does not require external criteria. However, because no criteria are referenced, additional research is needed to link resulting ranges to clinical decisions. Second, functional impairment is a common criterion for psychopathology thresholding, as it is fundamental to case definition and service eligibility determination. Its main limitation is that reliance on current impairment misses patients who are currently unimpaired but nonetheless at risk for future negative outcomes. Third, a prediction approach addresses this limitation and has been indispensable in other fields, such as internal medicine. Resulting risk ranges are essential for prevention and prognostication. However, they are not directly linked to treatment selection. Fourth, cost-benefit analysis of interventions can identify thresholds above which a given intervention is indicated by its cost-effectiveness. Unfortunately, this approach is nearly absent in psychopathology.

Ultimately, the choice of strategy depends on the intended application (eg, communicating psychopathology severity, deciding eligibility for care, making a prognosis, or selecting treatment), and each plays a unique role in improving psychiatric assessment. Psychopathology encompasses numerous constructs that require ranges. It seems likely that the best thresholding approach will differ between dimensions depending on proven utility of each: cost-benefit for constructs informative for treatment selection, prediction for those effective at forecasting, impairment for those strongly related to functioning, and statistical deviance for all others.

Practical next steps include collecting normative data for psychiatric assessments that lack these data and calibrating them accordingly. Functional impairment can feasibly be assessed in the same studies to determine which dimensions relate sufficiently to functioning to warrant impairment-based thresholds. Existing longitudinal data offer ample opportunities to identify dimensions useful for prognostication and explicate prediction-based thresholds. However, new follow-up studies will be needed to test constructs missing from existing longitudinal datasets. Likewise, economic evaluation may use both existing and targeted new data to identify dimensions and ranges that can guide selection of interventions. These approaches answer different questions and each is needed, but their relative contributions are currently unknown.

Generalizability of cutoffs is another consideration. Prevalence of outcomes differ by population, which may require a change of threshold to maintain the same characteristics in a new setting.¹⁸ Accordingly, the risk ranges outlined in Figure 2 apply to outpatients with a psychiatric disorder, but likely are not optimal in populations such as primary care or community youth. It is best to develop thresholds for multiple populations to ensure generalizability (eg, derivation of PCEs was stratified by demographics⁵⁴). Another challenge is underutilization of measurement-based care in many mental health settings. Without quantitative data, practitioners are unable to apply thresholds. Perceived low utility of assessment contributes to underutilization, and development of actionable ranges may change such views.

Thresholds cannot be chosen on statistical grounds alone, as no natural discontinuity in relationships between psychopathology and relevant criteria (eg, functional impairment) has been established to date.^{5,6} Thresholding creates an inescapable trade-off between

sensitivity and specificity (or PPV), and value judgment is necessary to weigh their benefits and harms.⁶² Optimally, the judgment should be made by a panel of experts, including patients, family advocates, policymakers, regulators, insurers, etc. Likewise, stakeholder input is essential in a cost-benefit analysis, particularly in assigning monetary value to QALYs. It falls on our professional organizations to convene such panels.

We focused primarily on clinical decisions, but scientists would likewise benefit from established ranges. Cutoffs can help further standardize sample selection and description, let alone highlight knowledge gaps in threshold development. This work may inform the *DSM-5-TR* and *ICD-11* revisions. Both manuals include dimensional measures (eg, personality disorder traits) that can benefit from actionable thresholds. Also, dimensions with established ranges are more likely to meet clinical utility requirement for inclusion in *DSM* and *ICD*.

Our goal was to outline options and considerations involved in developing purpose-based ranges for psychopathology dimensions. This issue has been largely neglected, but it is becoming more pressing as the field embraces the fundamentally continuous nature of psychopathology.^{1-5,9} dimensional nosologic models increase in prominence,¹²⁻¹⁶ and interest in personalized treatments grows (ie, *precision psychiatry*).⁷⁵ This is challenging for existing practice guidelines, as nearly all are tied to diagnostic categories. Moreover, relevant literature on psychopathology dimensions is limited. New research is needed to derive actionable ranges, and this work is feasible, given many excellent datasets that can be reanalyzed to determine thresholds. We hope that the present article facilitates these efforts. Development of purpose-based ranges could greatly improve the utility of psychiatric nosology,¹⁴ as it has done in other fields of medicine.

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